COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:	07/07/26	**DATE SUBMITTED:**	07/07/26
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ACTIVITY NO. 1.1 Creating Virtual Machines

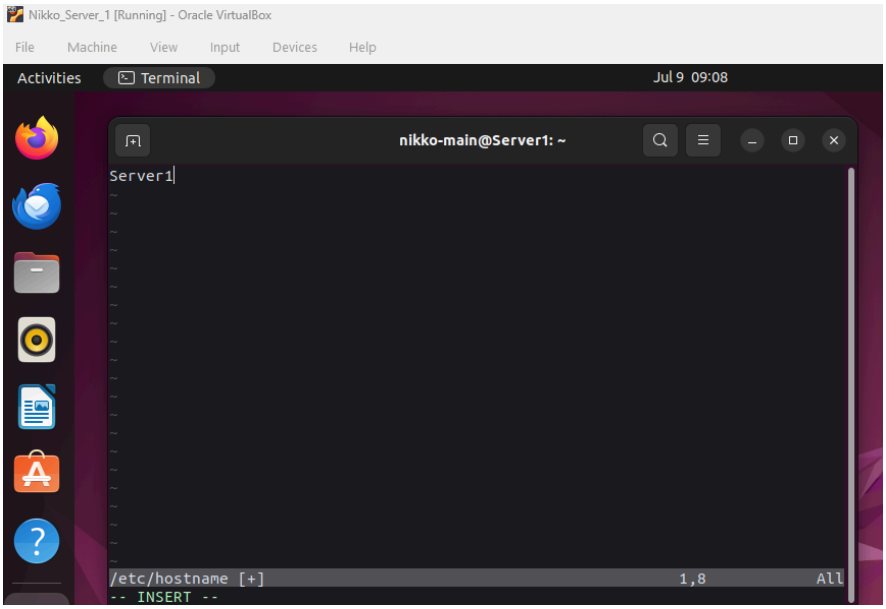
1. PROCEDURE OUTPUT

What procedures did you perform?

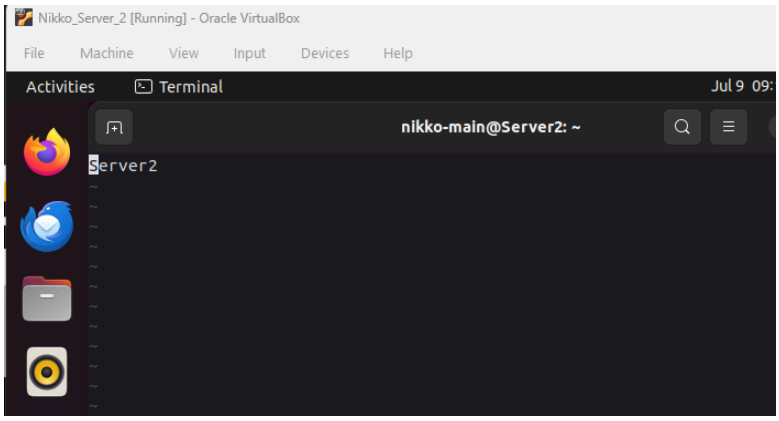
Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

1. Change the hostname using the command *sudo nano /etc/hostname*

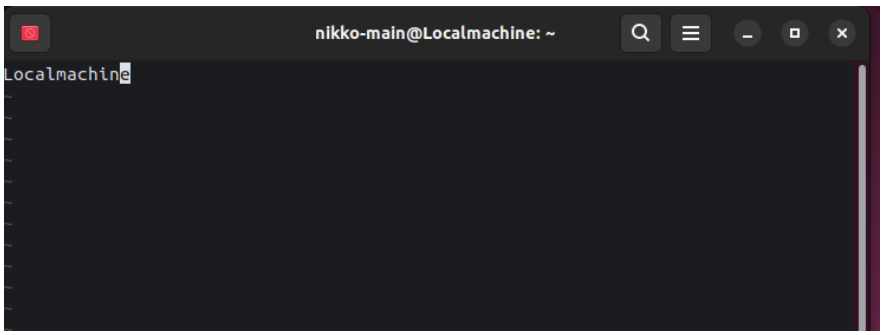
1.1 Use server1 for Server 1



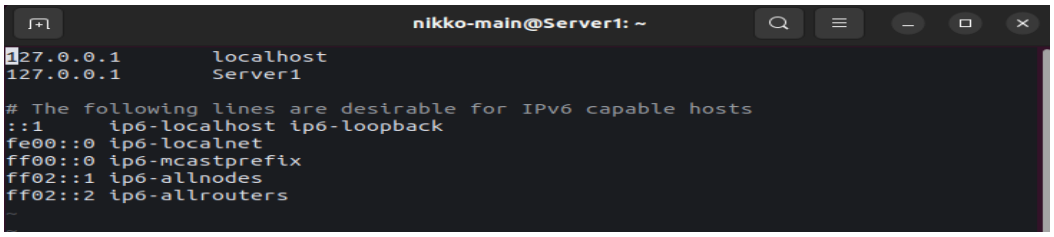
1.2 Use server2 for Server 2



1.3 Use workstation for the Local Machine



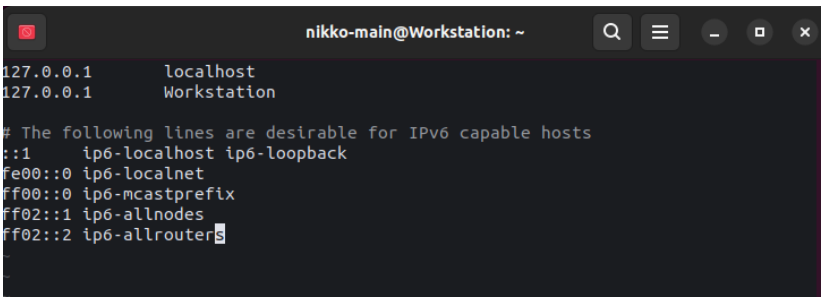
2. Edit the hosts using the command *sudo nano /etc/hosts.* Edit the second line.
2.1 Type 127.0.0.1 server 1 for Server 1



2.2 Type 127.0.0.1 server 2 for Server 2



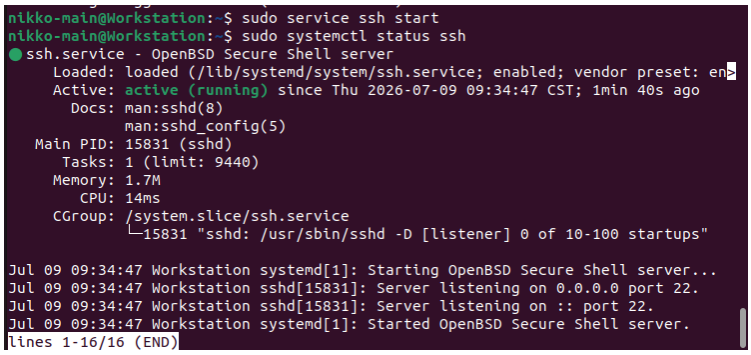
2.3 Type 127.0.0.1 workstation for the Local Machine



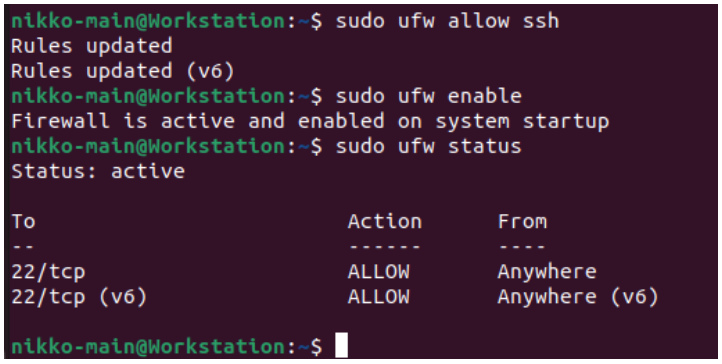
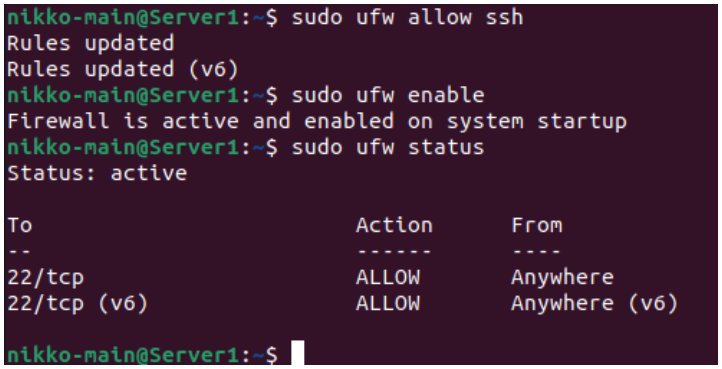
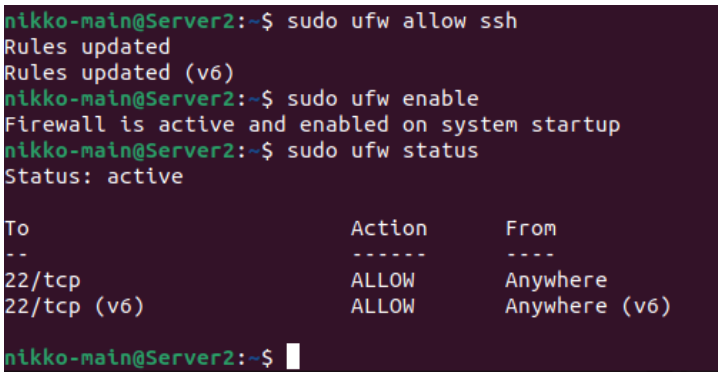
- **Task 1 Analysis:**
  The first task was straightforward. I configured the file containing the hostname and the IP addresses as required by the activity. Additionally, I used a different text editor, nvim, since it is easier for me to use. The /etc/hostname file sets the machine's name, while /etc/hosts is used for configuring each machine's network settings.

Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

1. Upgrade the packages by issuing the command *sudo apt update* and *sudo apt upgrade* respectively.
2. Install the SSH server using the command *sudo apt install openssh-server*.
3. Verify if the SSH service has started by issuing the following commands:
   3.1 *sudo service ssh start*
   3.2 *sudo systemctl status ssh*

	*Workstation*
	*Server 2*
	*Server 1*

4. Configure the firewall to all port 22 by issuing the following commands:
   4.1 *sudo ufw allow ssh*
   4.2 *sudo ufw enable*
   4.3 *sudo ufw status*

	*workstation*
	*Server 1*
	*Server 2*

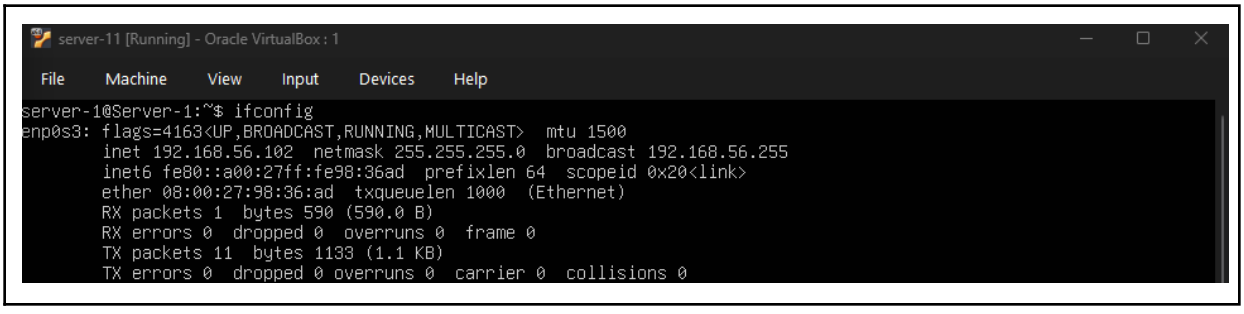
- **Task 2 Analysis:**

  The first step I took was to install the necessary packages for this task. I began by checking whether SSH was running using the command sudo systemctl status ssh, since this can vary between distributions. Next, I started the SSH service on my machine with sudo systemctl start ssh, and then checked the status again to confirm it was active. For the firewall setup, I used sudo ufw to define the rules, activated the firewall, and finally verified that the changes took effect.

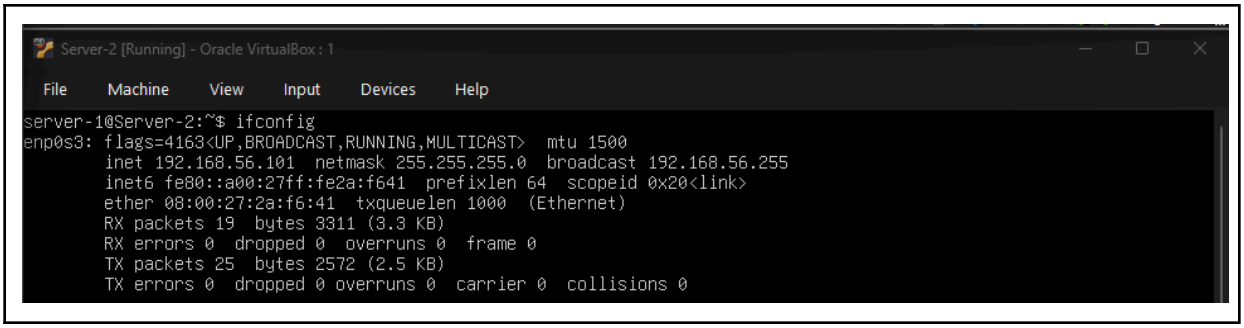
Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command *ifconfig* and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.

   1.1 Server 1 IP address: 192.168.56.102



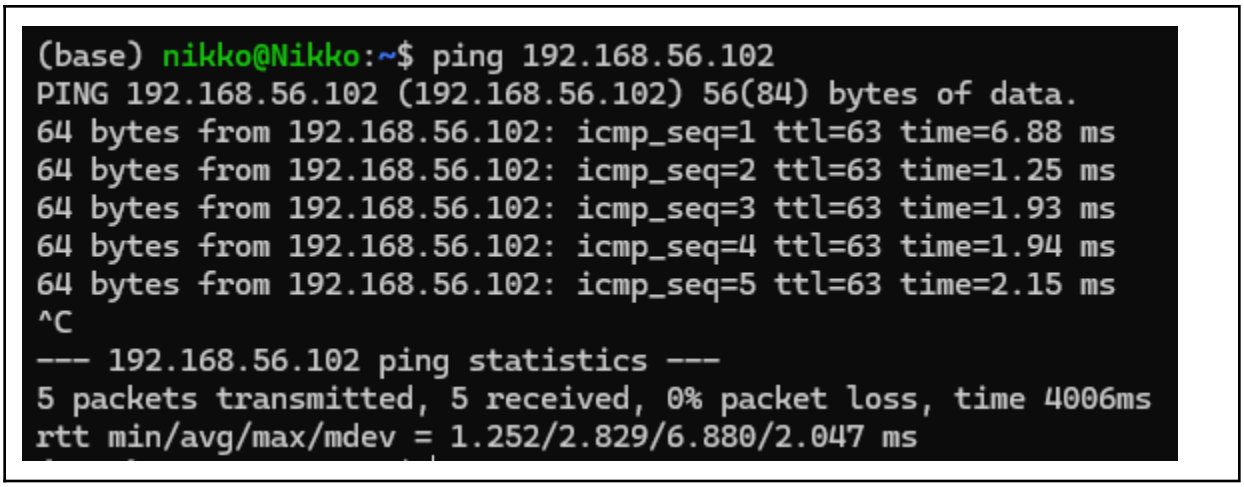
   1.2 Server 2 IP address: 192.168.56.118



1.3 WorkstationIP address: 192.168.56.117



2. Make sure that they can ping each other.
   2.1 Connectivity test for Local Machine 1 to Server 1:
   ✅ Successful
   Not Successful



2.2 Connectivity test for Local Machine 1 to Server 2:

✅ Successful

☐ Not Successful

```
(base) nikko@Nikko:~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=63 time=3.75 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=63 time=1.80 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=63 time=2.00 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=63 time=1.73 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=63 time=2.44 ms
^C
--- 192.168.56.101 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.734/2.344/3.750/0.745 ms
```

2.3 Connectivity test for Server 1 to Server 2:

✅ Successful

☐ Not Successful

```
server-1@Server-1:~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=4.36 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.08 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=1.45 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.23 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=1.62 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=1.18 ms
^C
--- 192.168.56.101 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5124ms
rtt min/avg/max/mdev = 1.075/1.818/4.360/1.150 ms
```

- **Task 3 Analysis:**

  I just verify if the connection was stablished successfully, using ping and the ip addresses. SInce the ip address is different when the network is in NAT I changed it to host only adapter, but the problem is that I can install anything when using that network. So my solution was to use 2 adapters, Nat and host only.

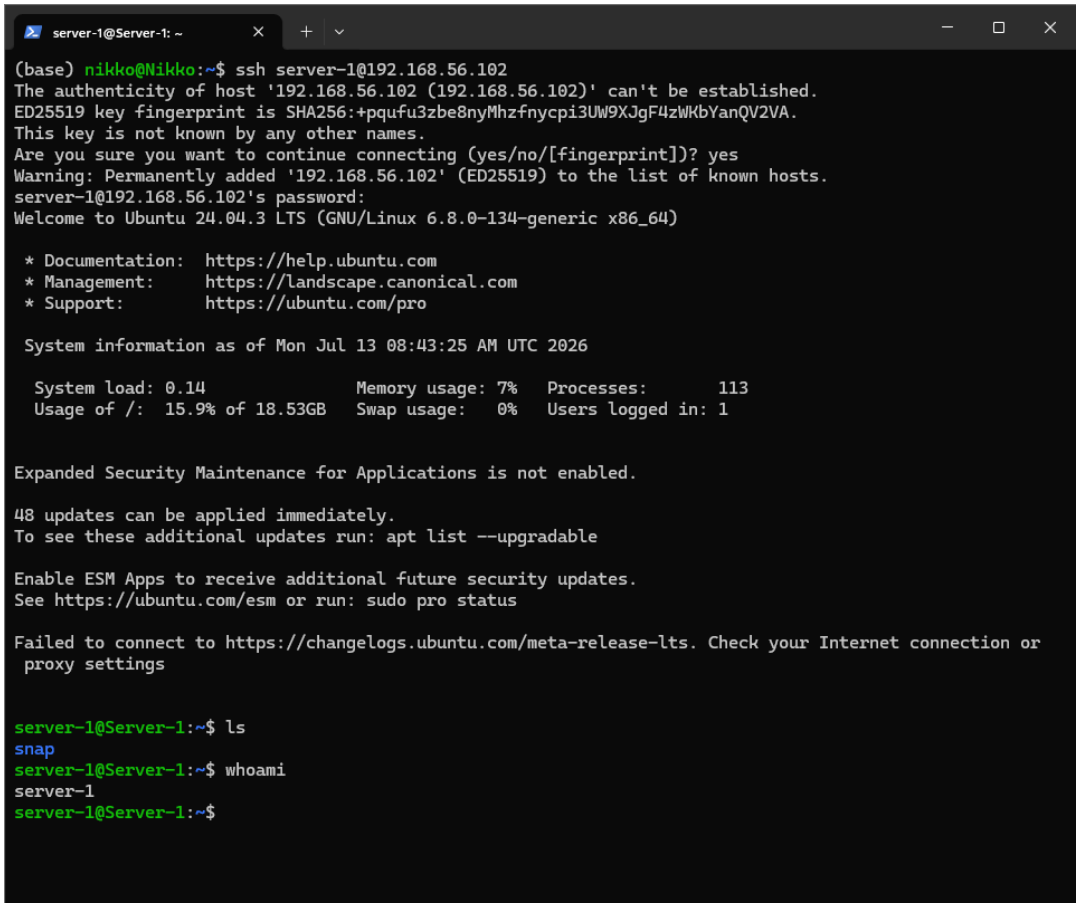
Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the Local Machine, issue the following commands:

1.1 ssh username@ip_address_server1 for example, *ssh jvtaylar@192.168.56.120*

1.2 Enter the password for server 1 when prompted

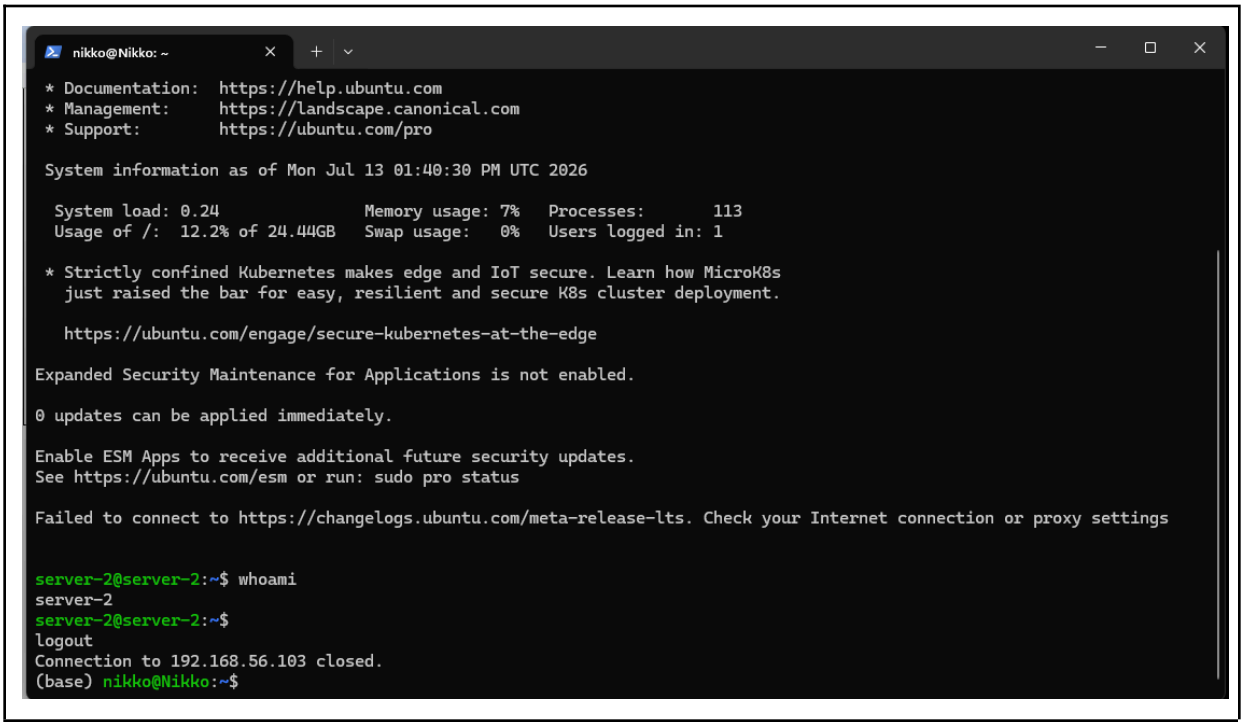
1.3 Verify that you are in server 1. The user should be in this format user@server1. For example, *jvtaylar@server1*



2. Logout of Server 1 by issuing the command *control + D.*



3. Do the same for Server 2.



4. Edit the hosts of the Local Machine by issuing the command *sudo nano /etc/hosts.* Below all texts type the following:

4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)
4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)
4.3 Save the file and exit.
5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do *ssh jvtaylar@server1*. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2.

Server 1

```
(base) nikko@Nikko:~$ ssh server-1@server1
The authenticity of host 'server1 (192.168.56.102)' can't be established.
ED25519 key fingerprint is SHA256:+pqufu3zbe8nyMhzfnycpi3UW9XJgF4zWKbYanQV2VA.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server1' (ED25519) to the list of known hosts.
server-1@server1's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Jul 13 02:25:29 PM UTC 2026

  System load: 0.25               Memory usage: 6%   Processes:       112
  Usage of /:  16.0% of 18.53GB   Swap usage:   0%   Users logged in: 1


Expanded Security Maintenance for Applications is not enabled.

48 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or
 proxy settings


Last login: Mon Jul 13 11:24:35 2026 from 192.168.56.1
server-1@Server-1:~$
logout
Connection to server1 closed.
(base) nikko@Nikko:~$ DX
```

Server 2

- **Task 4 Analysis:**

  Using ssh to remotely access the other machine. Then configure the file to remember the ip address and the hostname which will make it much more easier to access.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:
Answer the following:
How are we able to use the hostname instead of IP address in SSH commands?

- Basically, the file is like a DNS which will be used to remember anything that was configured like earlier. In task 4 it was used for automatically accessing the other machine. I just need to use the hostname and the machine name. I don't have to use the ip address.

How secured is SSH?

- Since ssh uses a cryptography process encrypt and decrypt the data.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini, grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction, explanation of concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.



I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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